1.1 McCulloch y Pitts

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1 Caso Práctico: Neurona de McCulloch y Pitts

1.1 1. Implementación de la MPNeuron

In [2]:
```
import numpy as np

class MPNeuron:

    def __init__(self):
        self.threshold = None

    def model(self, x):
        # input: [1, 0, 1, 0] [x1, x2, .., xn]
        z = sum(x)
        return (z >= self.threshold)

    def predict(self, X):
        # input: [[1, 0, 1, 0], [1, 0, 1, 1]]
        Y = []
        for x in X:
            result = self.model(x)
            Y.append(result)
        return np.array(Y)
```

In [3]:
```python
# Instanciamos la neurona
mp_neuron = MPNeuron()
```

In [4]:
```python
# Establecemos un threshold
mp_neuron.threshold = 4
```

In [5]:
```python
# Evaluamos diferentes casos de uso
mp_neuron.predict([[1, 0, 0], [1, 1, 1, 1], [1, 1, 1, 0]])
```

Out[5]:
```
array([False,  True, False])
```

5. Github

https://github.com/Erick-305/machine-learning.git